

SPECIAL THEORY OF RELATIVITY

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ABSTRACT. Quantum Field Theory (QFT) provides the essential theoretical framework that unifies quantum mechanics and special relativity. These notes explore the foundational concepts and mathematical machinery required to describe systems with infinite degrees of freedom, where particles are understood as localized excitations of underlying fields. Beginning with the transition from classical to quantum field formalisms, the text develops the canonical and path integral methods of quantization. Core topics include the analysis of free and interacting fields, the derivation of scattering amplitudes, and the implementation of diagrammatic perturbation theory. Symmetries, conservation laws, and the techniques necessary to handle regularized infinities are examined systematically. Ultimately, this material establishes the universal language used to analyze complex, multi-particle interactions across both high-energy and many-body physics.

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1. SPECIAL RELATIVITY: BASIC CONCEPTS

1.1. The Reference Frame

The reference frame sits one step below the fully relativistic regime and is a concept we have encountered extensively in classical mechanics.

Definition 1.1. A reference frame is a set of spatial coordinates used to specify *where* and *when* an event occurs.

A complete reference frame is not described by three coordinates, but by four.

$$F_r = (x, y, z, t) \tag{1}$$

This four-coordinate description may be characterised as “pre-relativistic,” as it is the framework we have already worked with. With that established, what is relativity?

Definition 1.2. Relativity states that for all reference frames related to one another by uniform motion, the laws of physics remain the same — or more precisely, they remain *invariant*.

Notably, the principle of relativity was not introduced by Einstein; it predates him considerably. Einstein’s key contribution was to elevate the speed of light to the status of a law of physics.

$$c = 186,000 \text{ miles/sec} \tag{2}$$

$$c = 3.00 \cdot 10^8 \text{ m/s} \tag{3}$$

The consequence of this addition is striking. If the principle of relativity demands that all laws of physics are the same in every reference frame, and the speed of light is a law of physics, then the speed of light must be the same in every reference frame. This is a genuinely puzzling result, and it is precisely what we now set out to understand.

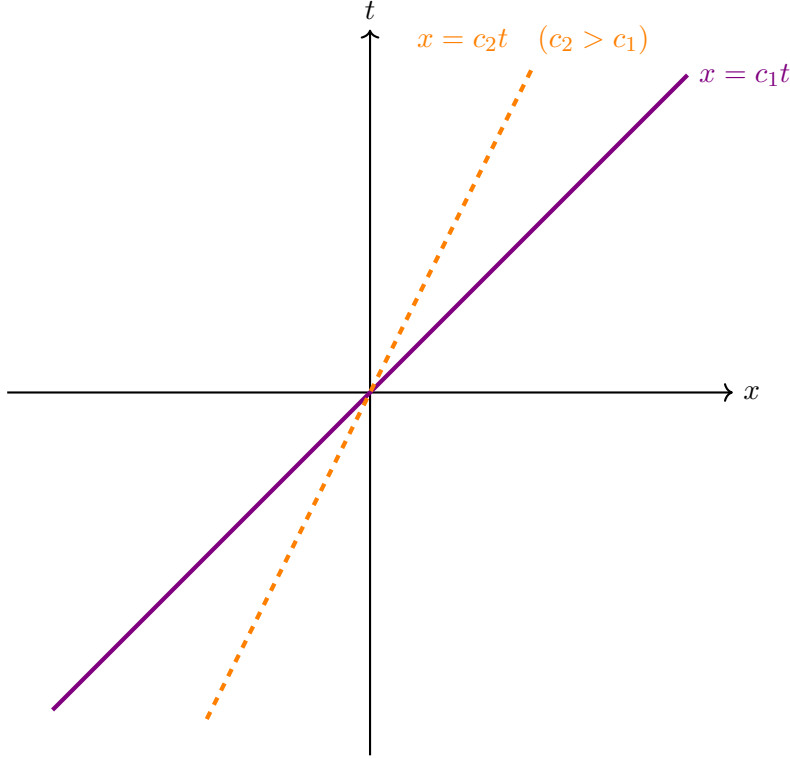


Figure 1: Spacetime diagram comparing the original slope to a steeper slope as c grows.

To simplify the discussion, we restrict space to one dimension. A light ray emitted from the origin travels according to $x = ct$. If c is taken to be its actual physical value, the worldline is so steep on a spacetime diagram that it becomes nearly indistinguishable from the t -axis. This is made explicit by Eq. (4), which rearranges to give Eq. (5):

$$x = ct \quad (4)$$

$$t = \frac{1}{c}x \quad (5)$$

Consider, relative to the frame in Fig. 1, a helpful and intuitive example: a stationary frame as just described, and a second frame moving with uniform velocity relative to it. Concretely, think of Fig. 1 as representing yourself standing at the side of a road, with this new frame corresponding to a friend driving past in a car.

From your reference frame, at time $t = 0$ your friend passes you travelling along $x = vt$, where we use v because your friend is not travelling at the speed of light. From your friend's perspective, however, they are stationary inside their car: they have not moved from the driver's seat, so in their frame $x' = 0$ at all times.

At some point, the two of you must reconcile your descriptions of events. To do this, we need to express x' in terms of x , which requires us to employ the concept of **linearity**.

In flat spacetime, space and time are assumed to be homogeneous (uniform). This requires that straight worldlines in one frame map to straight worldlines in another. The transformation must therefore be a linear combination:

$$x' = Ax + Bt \quad (6)$$

where A and B are constant coefficients to be determined.

We impose the physical constraint that the origin of the moving frame ($x' = 0$) travels at constant velocity v relative to the stationary frame. In the stationary observer's coordinates, this path is:

$$x = vt \quad (7)$$

Substituting this boundary condition ($x' = 0$ when Eq. (7) holds) into Eq. (6) yields:

$$0 = A(vt) + Bt \quad (8)$$

$$0 = (Av + B)t \quad (9)$$

For this equation to hold true at all times ($t \neq 0$), the expression inside the parentheses must vanish:

$$Av + B = 0 \implies B = -Av \quad (10)$$

Replacing B with $-Av$ in Eq. (6) gives:

$$x' = Ax + (-Av)t \quad (11)$$

Factoring out the shared constant A simplifies the structure to:

$$x' = A(x - vt) \quad (12)$$

As a side note, we can determine the value of A that satisfies this relation. Recall that the two observers must agree on a common description of events. In classical mechanics, this is a consequence of space being absolute: both frames must agree on the scale of a unit distance. Requiring $x' = x$ at $t = 0$ gives:

$$x' = Ax \quad (13)$$

For this to hold, we must have $A \equiv 1$. In summary, we have recovered the **Galilean transformation**:

$$x' = x - vt \quad (14)$$

$$t' = t \quad (15)$$

Now consider the beam of light $x = ct$. How does it transform between frames? Substituting Eq. (4) into Eq. (14):

$$x' = ct - vt = (c - v)t = (c - v)t' \quad (16)$$

There is something clearly wrong with Eq. (16). Einstein postulated that the speed of light is the same in every reference frame, yet in the moving frame the light appears to travel at $c - v$, not c .

The problem is compounded when considering light travelling in the opposite direction, $x = -ct$. Substituting into $x' = x - vt$ gives $x' = -(c + v)t = -(c + v)t'$, implying the moving observer measures a speed of light *greater* than c . Both results violate Einstein's postulate, and experiment confirms that the speed of light is indeed the same in all cases. So the question becomes: where does the mathematics break down?

The issue lies with the transformation $t = t'$. If two observers synchronise their clocks at the moment they pass one another and then separate, will their clocks continue to agree? This is the **assumption of simultaneity**, and it turns out to be unjustified.

Simultaneity is **frame dependent**. To begin to understand why, we first need the concept of **clock synchronisation**. Einstein's operational definition goes as follows: two observers wish to synchronise their clocks. Each sends a pulse of light toward a mutually agreed midpoint. If the pulses arrive at the midpoint simultaneously, the clocks are deemed synchronised.

Now introduce a third observer moving past the midpoint. From their perspective, the two pulses may not arrive simultaneously, because they themselves have moved past the midpoint during the travel time of the pulses. They would conclude the clocks are not synchronised. Rather than calling this "unfair," the correct takeaway is that **synchronicity is frame-dependent**: different observers genuinely disagree about whether two events are simultaneous.

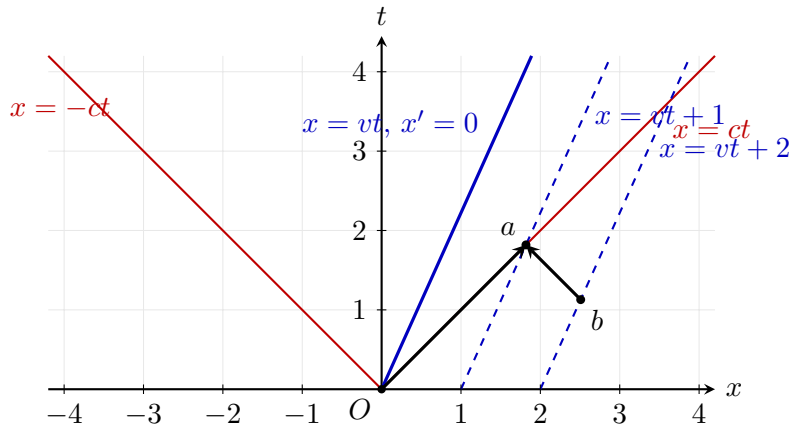
To make this concrete, consider a system consisting of a light ray and a collection of moving observers spaced at equal intervals. Recall that a forward-travelling beam of light satisfies Eq. (17),

$$x = ct \quad (17)$$

while a backward-travelling beam satisfies

$$x = -ct \quad (18)$$

We now superimpose the observers' worldlines onto this diagram to determine the location of the t' axis.



To re-state the setup: each integer position along the x -axis is occupied by an observer. The observer at the origin O and the observer at position 2 each send a light signal toward the observer at position 1, with the aim of having both signals arrive simultaneously. Our goal is to determine when that arrival event occurs. These signals are indicated by the black lines in Fig. 2.

From Fig. 2, we see that in the stationary frame, light moves along 45° lines. For the moving observer, $t' = 0$ at the arrival event, since they perceive both pulses arriving simultaneously. It remains only to determine the coordinates of points a and b .

Point a is the simpler of the two, so we begin there. It lies at the intersection of two worldlines, so we solve the corresponding system of equations. From Fig. 2, point a lies on:

$$x = t \quad (19)$$

and

$$x = vt + 1 \quad (20)$$

Note that we set the speed of light c equal to 1. Solving this system of equations,

$$t = vt + 1 \quad (21)$$

$$t - vt = 1 \quad (22)$$

$$t(1 - v) = 1 \quad (23)$$

$$\boxed{t_a = \frac{1}{1-v}} \quad (24)$$

Using the fact that $x = t$, we then know the x -coordinate at point a as:

$$\boxed{x_a = t = \frac{1}{1-v}} \quad (25)$$

Now we turn to $\overline{ab}$. Along a 45° line directed downward-right (or equivalently upward-left), the quantity $x + t$ is constant. To find this constant, we use the fact that point a lies on this line:

$$(x + t)_a = 2 \cdot \frac{1}{(1-v)} \quad (26)$$

To find point b , we intersect $\overline{ab}$ with the worldline of the second observer, Eq. (28). Substituting,

$$x + t = \frac{2}{1-v} \quad (27)$$

and

$$x = vt + 2 \quad (28)$$

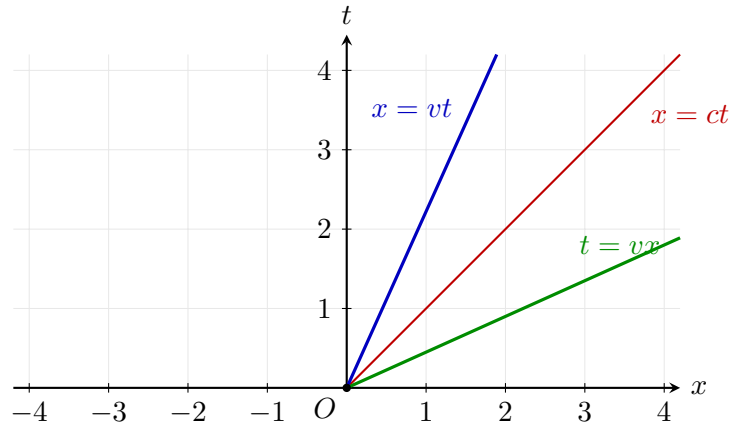
Substituting Eq. (28) into Eq. (27):

$$(vt + 2) + t = \frac{2}{1-v} \quad (29)$$

$$\boxed{t_b = \frac{2v}{1-v^2}} \quad (30)$$

Re-arranging $x = \frac{2}{1-v} - t$ and substitution of t_b yields:

$$\boxed{x_b = \frac{2}{1-v} - \frac{2v}{1-v^2} = \frac{2}{1-v^2}} \quad (31)$$



The green line in Fig. 3 represents the ratio t_b/x_b . Notice that the angle between the t -axis and the x -axis is the same as the angle between the x' -axis and the t -axis. This is precisely the geometric content of Einstein's observation about synchronicity.

We can now say something more precise about the relationships between x, t, x', t' . Revisiting Eq. (14), we note that this equation may be incorrect by some factor depending on velocity, which we denote $f(v)$:

$$x' = (x - vt)f(v) \quad (32)$$

Applying the same reasoning to t' : we found that $x' = 0$ corresponds to $x = vt$, and by a parallel argument $t' = 0$ corresponds to $t = vx$. We therefore introduce a corrective factor $g(v)$ for the time transformation:

$$t' = (t - vx)g(v) \quad (33)$$

We can apply Einstein's postulate — that the speed of light is $c = 1$ in every frame — to constrain these factors. In the stationary frame, the light cone satisfies $x = \pm t$; the same must hold in the moving frame, so $x' = \pm t'$ whenever $x = \pm t$.

To ensure this, recall Eqs. (34) and (33):

$$x' = (x - vt)f(v) \quad (34)$$

$$t' = (t - vx)g(v) \quad (35)$$

When $x = t$, the expressions $(x - vt)$ and $(t - vx)$ are identical, so requiring $x' = t'$ forces:

$$f(v) = g(v) \quad (36)$$

The remaining task is to determine this common function. To do so, we write the inverse relations:

$$x = (x' + vt')f(v) \quad (37)$$

$$t = (t' + vx')g(v) \quad (38)$$

1.2. Lorentz Transformations

To verify that our expressions for x, t, x', t' are mutually consistent, we substitute the formulas for x' and t' back into the expressions for x and t . The result should reproduce the original coordinates identically. This consistency requirement is what fixes the function of velocity.

Substituting Eqs. (34) and (33) into Eqs. (37) and (38) and expanding:

$$x = [(x - vt)f(v) + v(t - vx)f(v)]f(v) \quad (39)$$

$$= [(x - vt)f^2(v) + v(t - vx)f^2(v)] \quad (40)$$

$$= xf^2(v) - vtf^2(v) + vtf^2(v) - v^2xf^2(v) \quad (41)$$

$$= xf^2(v) - v^2xf^2(v) \quad (42)$$

$$= xf^2(v)(1 - v^2) \quad (43)$$

$$= x \quad (44)$$

This tells us that the function of velocity $f(v)$ must be:

$$f = \frac{1}{\sqrt{1 - v^2}} \quad (45)$$

Substituting this into Eqs. (34) and (33) yields the final transformation equations:

$$x' = \frac{(x - vt)}{\sqrt{1 - v^2}} \quad (46)$$

$$t' = \frac{(t - vx)}{\sqrt{1 - v^2}} \quad (47)$$

These are the **Lorentz transformations**. They guarantee that the speed of light equals 1 in every inertial frame of reference. If one works in units where $c \neq 1$, the equations must be rescaled to remain geometrically consistent.

Definition 1.3. Lorentz transformations are a set of equations that relate the spacetime coordinates of an event as recorded by one inertial observer to those recorded by another.

Examining Eq. (46) more carefully: the denominator is dimensionless only when v is measured in units of c . To restore explicit factors of c and make the units consistent, we write:

$$x' = \frac{(x - vt)}{\sqrt{1 - \frac{v^2}{c^2}}} \quad (48)$$

$$t' = \frac{(t - \frac{v}{c^2}x)}{\sqrt{1 - \frac{v^2}{c^2}}} \quad (49)$$

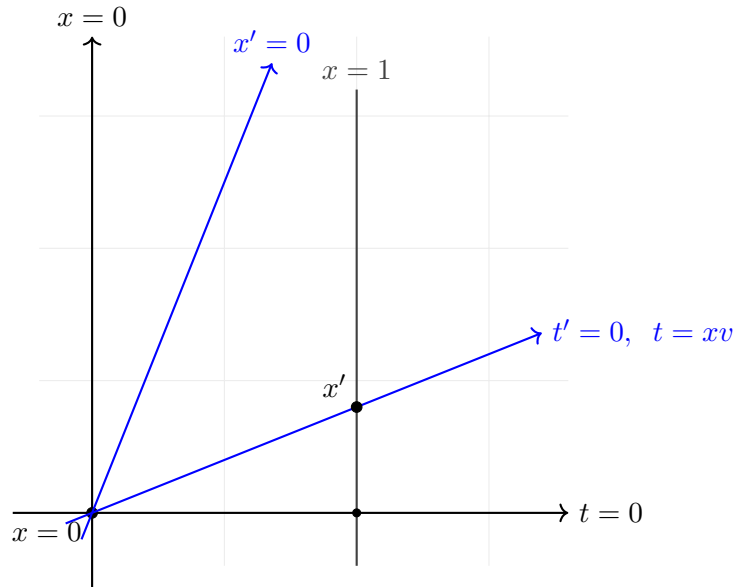
The dimensionless velocity ratio appearing in the denominator is conventionally written as:

$$\frac{v^2}{c^2} = \beta \quad (50)$$

These transformations are restricted to the x - t plane. The transverse directions are unaffected: $y' = y$ and $z' = z$.

1.3. Time Dilation

Definition 1.4. Time dilation is the phenomenon where a clock moving at a constant velocity v relative to an inertial observer is measured to tick slower than a clock at rest in that observer's reference frame.



To understand the concept of time dilation, the graph above describes the following situation.

1. **The Meter Stick is at Rest in the Ground Frame (S):** The meter stick is not moving; it rests on the ground along the horizontal x -axis, spanning from $x = 0$ to $x = 1$. Because it remains stationary over time, its back end stays on the vertical axis ($x = 0$) and its front tip remains at $x = 1$ for all values of t . This traces the vertical worldline labeled $x = 1$ on the whiteboard.
2. **The Moving Observer (S') Travels Past:** A moving observer (S') travels along the x -axis at a constant velocity v . Their slice of simultaneous space (the events they perceive as happening at time zero) is represented by their tilted spatial axis, labeled $t' = 0$. In coordinate terms of the ground frame, this axis follows the linear path:

$$t = v \cdot x \quad (51)$$

3. **The Objective (x'):** The goal is to determine what coordinate value x' the moving observer assigns to the front tip of the ground's meter stick at the instant $t' = 0$. Geometrically, this value is found at the intersection point marked x' on the diagram, where the stick's vertical worldline ($x = 1$) cuts through the moving observer's tilted space axis ($t = vx$).

To find the value of x' , recall the Lorentz transformations with $c = 1$.

$$x' = \frac{(x - vt)}{\sqrt{1 - v^2}}$$

$$t' = \frac{(t - vx)}{\sqrt{1 - v^2}}$$

Substituting $t = vx$ and $x = 1$ into the expression for x' :

$$x' = \frac{(x - vt)}{\sqrt{1 - v^2}} \quad (52)$$

$$= \frac{(x - v(vx))}{\sqrt{1 - v^2}} \quad (53)$$

$$= \frac{x - xv^2}{\sqrt{1 - v^2}} \quad (54)$$

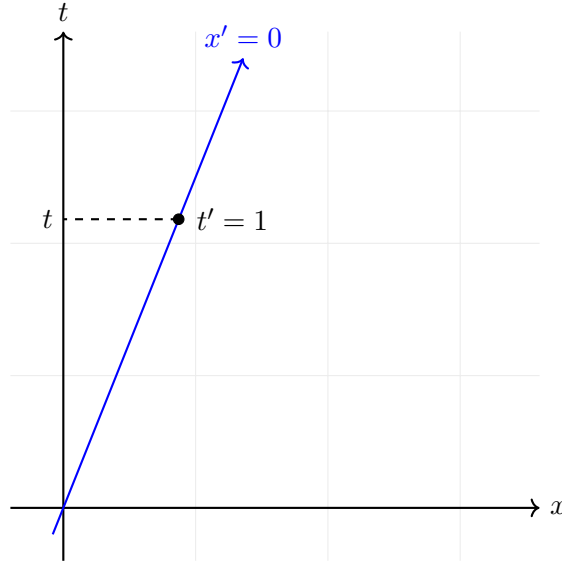
$$= \frac{x(1 - v^2)}{\sqrt{1 - v^2}} \quad (55)$$

$$= \frac{(1 - v^2)}{\sqrt{1 - v^2}} \quad (56)$$

$$= \boxed{\sqrt{1 - v^2}} \quad (57)$$

$$(58)$$

This result shows that length, like time, is not an absolute quantity — it is frame-dependent. To now examine the same effect for a moving clock:



The diagram above describes the following physical setup:

1. **The Clock Travels with the Moving Observer (S'):** Because the clock is at rest inside the moving frame, its spatial coordinate in that frame never changes:

$$x' = 0 \quad (59)$$

This traces the tilted blue worldline labelled $x' = 0$ in the diagram.

2. **The First Tick ($t' = 1$):** We identify the event at which the moving clock records exactly one unit of elapsed time, marking the point $t' = 1$.
3. **The Objective (t):** The goal is to determine what coordinate time t the stationary observer reads when the moving clock ticks at $t' = 1$. Geometrically, since both observers are examining the same event, we project a horizontal dashed line from that event to the vertical t -axis.

To compute this value of t , we apply the Lorentz transformation equations directly with our constraints.

First, substitute the spatial constraint $x' = 0$ into the position transformation to obtain the relationship between x and t in the stationary frame:

$$x' = \frac{x - vt}{\sqrt{1 - v^2}} \quad (60)$$

$$0 = \frac{x - vt}{\sqrt{1 - v^2}} \quad (61)$$

$$0 = x - vt \quad (62)$$

$$x = v \cdot t \quad (63)$$

Substituting this result ($x = vt$) together with the time constraint ($t' = 1$) into the Lorentz

transformation for time:

$$t' = \frac{t - vx}{\sqrt{1 - v^2}} \quad (64)$$

$$1 = \frac{t - v(vt)}{\sqrt{1 - v^2}} \quad (65)$$

$$1 = \frac{t - vt^2}{\sqrt{1 - v^2}} \quad (66)$$

$$1 = \frac{t(1 - v^2)}{\sqrt{1 - v^2}} \quad (67)$$

Using the algebraic identity $\frac{A}{\sqrt{A}} = \sqrt{A}$ where $A = (1 - v^2)$, the equation simplifies directly to:

$$1 = t \cdot \sqrt{1 - v^2} \quad (68)$$

$$t = \boxed{\frac{1}{\sqrt{1 - v^2}}} \quad (69)$$

Since $\sqrt{1 - v^2} < 1$ for any $v > 0$, its reciprocal satisfies $t > 1$. This confirms that when the moving observer has experienced exactly one second on their clock ($t' = 1$), the stationary observer measures a longer elapsed time ($t > 1$). From the stationary frame's perspective, the moving clock runs slow. This is the origin of the **twin paradox**.

1.4. Twin Paradox

From Fig. 5, the twin paradox arises by reversing the motion of the travelling twin and examining how the elapsed time differs between the two reference frames.

According to the time dilation result derived above, Twin A observes Twin B's clock as running slow, so Twin B ages more slowly. When Twin B returns to Earth, they will be younger than Twin A.

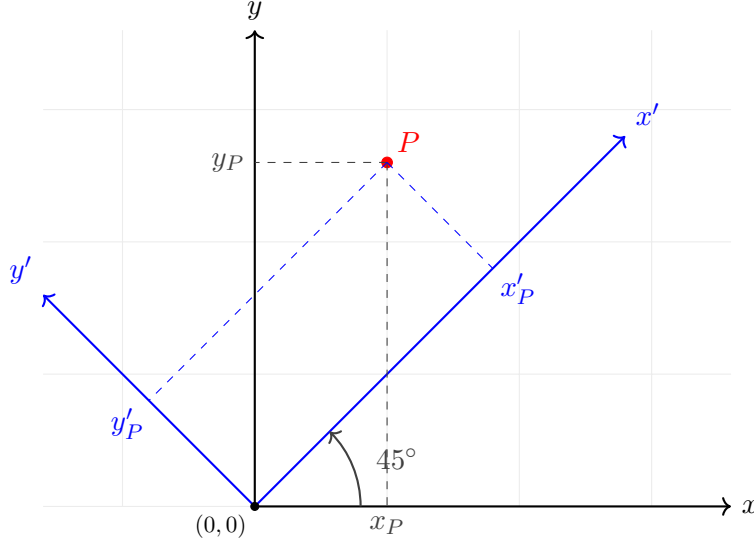
The "Paradox". The apparent paradox stems from the core principle of relativity: motion is relative. From Twin B's perspective inside the rocket ship, it is Twin A (and the Earth) that departed and returned. Twin B could therefore argue that Twin A's clock was running slow, and that Twin A should be the younger one upon reunion. Since both twins cannot simultaneously be younger than the other, this seems to be a logical contradiction.

Resolving the Paradox. The paradox is resolved by recognising that the two twins' journeys are **not symmetrical**.

1. **Twin A** stays inside a single inertial reference frame for the entire duration of the trip.
2. **Twin B** must fire their engines to speed up, slow down, turn around, and decelerate to a stop back on Earth. This means Twin B must **accelerate**, breaking the symmetry.

This leads us into the discussion of Euclidean Geometry.

1.5. Euclidean Geometry



Coordinate systems in Euclidean space E^3 are related by rotations. As seen in Fig. 6, a point in space is not described by the same coordinates in the two systems.

The one quantity that remains constant is the distance from the origin: it is invariant under a change of coordinates, as guaranteed by Pythagoras' theorem:

$$x^2 + y^2 = x'^2 + y'^2 \quad (70)$$

The natural question is whether an analogous invariant exists for Lorentz transformations. A first guess might be:

$$t'^2 + x'^2 = t^2 + x^2 \quad (71)$$

Substituting eqs. (46), (47) into eq.(71). we obtain:

$$\left(\frac{t - vx}{\sqrt{1 - v^2}} \right)^2 + \left(\frac{x - vt}{\sqrt{1 - v^2}} \right)^2 = t^2 + x^2 \quad (72)$$

Squaring the numerators and denominators explicitly:

$$\frac{(t - vx)^2}{1 - v^2} + \frac{(x - vt)^2}{1 - v^2} = t^2 + x^2 \quad (73)$$

Since the denominators match, we combine the numerators over a single denominator:

$$\frac{(t - vx)^2 + (x - vt)^2}{1 - v^2} = t^2 + x^2 \quad (74)$$

Expanding the squared binomials in the numerator:

$$(t - vx)^2 = t^2 - 2vxt + v^2x^2 \quad (75)$$

$$(x - vt)^2 = x^2 - 2vxt + v^2t^2 \quad (76)$$

Substitute these expanded components back into the numerator:

$$\frac{(t^2 - 2vxt + v^2x^2) + (x^2 - 2vxt + v^2t^2)}{1 - v^2} = t^2 + x^2 \quad (77)$$

Combine the shared variable terms (t^2 and x^2) and the cross-terms (vxt) in the numerator:

$$\frac{t^2(1+v^2) + x^2(1+v^2) - 4vxt}{1-v^2} = t^2 + x^2 \quad (78)$$

Factoring out $(1+v^2)$ from the first two terms:

$$\frac{(1+v^2)(t^2+x^2) - 4vxt}{1-v^2} = t^2 + x^2 \quad (79)$$

Multiply both sides of the equation by the denominator $(1-v^2)$ to isolate the terms:

$$(1+v^2)(t^2+x^2) - 4vxt = (1-v^2)(t^2+x^2) \quad (80)$$

For Eq. (80) to hold for all independent values of x and t , the coefficients on both sides must match, which they do not under Eq. (71). However, the analogous relation with a sign difference does work:

$$t'^2 - x'^2 = t^2 - x^2 \quad (81)$$

Substituting eqs. (46), (47) into eq.(81), we obtain:

$$\left(\frac{t-vx}{\sqrt{1-v^2}} \right)^2 - \left(\frac{x-vt}{\sqrt{1-v^2}} \right)^2 = t^2 - x^2 \quad (82)$$

Squaring the numerators and denominators explicitly:

$$\frac{(t-vx)^2}{1-v^2} - \frac{(x-vt)^2}{1-v^2} = t^2 - x^2 \quad (83)$$

Since the denominators match, we combine the numerators over a single denominator:

$$\frac{(t-vx)^2 - (x-vt)^2}{1-v^2} = t^2 - x^2 \quad (84)$$

Expanding the squared binomials in the numerator:

$$(t-vx)^2 = t^2 - 2vxt + v^2x^2 \quad (85)$$

$$(x-vt)^2 = x^2 - 2vxt + v^2t^2 \quad (86)$$

Substitute these expanded components back into the numerator:

$$\frac{(t^2 - 2vxt + v^2x^2) - (x^2 - 2vxt + v^2t^2)}{1-v^2} = t^2 - x^2 \quad (87)$$

Combine the shared variable terms (t^2 and x^2) and the cross-terms (vxt) in the numerator:

$$\frac{t^2(1-v^2) - x^2(1-v^2)}{1-v^2} = t^2 - x^2 \quad (88)$$

Factoring out $(1-v^2)$ from both terms:

$$\frac{(1-v^2)(t^2-x^2)}{1-v^2} = t^2 - x^2 \quad (89)$$

Canceling the common factor $(1 - v^2)$ from the numerator and denominator yields the identity:

$$t^2 - x^2 = t'^2 - x'^2 \quad (90)$$

The left-hand side of Eq. (90) is the invariant quantity. Along the worldline of a moving observer we have $x' = 0$, so Eq. (81) reduces to:

$$\boxed{\tau^2 = t^2 - x^2} \quad (91)$$

Eq. (91) describes the elapsed time along the worldline of a moving clock. This is what we define to be the **proper time** — or more precisely, the proper time squared.

Definition 1.5. Proper time is the time interval measured by a single clock that accompanies an observer through two consecutive events.

This result also demonstrates that Euclidean geometry is not the appropriate framework for spacetime.

2. VECTORS AND SPIN IN 3-DIMENSIONAL SPACE AND IN RELATION TO SPECIAL RELATIVITY

2.1. Notes on Light

As the title of this subsection suggests, terminology can be illuminating. Here we take a brief detour to examine some properties of light — a useful reminder that even familiar concepts often contain more depth than first appears.

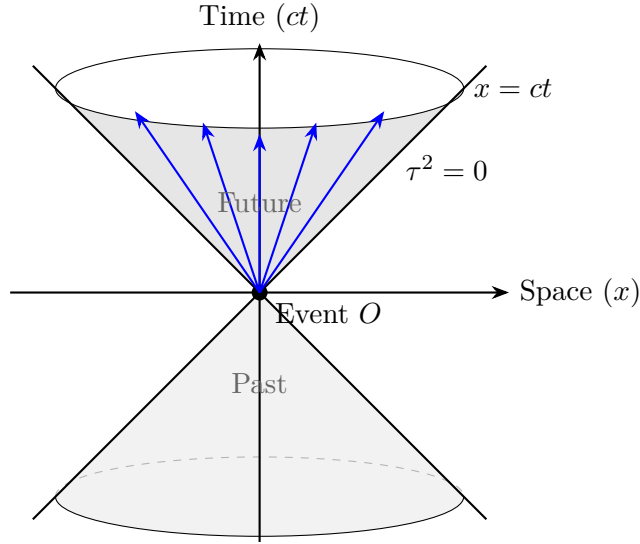


Figure 2: Minkowski Spacetime Diagram showing the Light Cone structures.

2.2. The Geometry of Light Cones in Minkowski Spacetime

The causal structure of special relativity is elegantly captured by the concept of the light cone, which partitions four-dimensional spacetime based on the invariant interval $ds^2 = -c^2 dt^2 + dx^2 + dy^2 + dz^2$.

Definition 2.1 (The Light Cone). The light cone of a given event O (chosen as the origin) is the three-dimensional hypersurface in four-dimensional spacetime swept out by all possible trajectories

of light rays passing through that event. Mathematically, it consists of all spacetime points that satisfy the null interval condition:

$$s^2 = -c^2t^2 + \vec{x}^2 = 0 \implies |\vec{x}| = \pm ct \quad (92)$$

The boundaries of the cone represent the absolute cosmic speed limit.

Definition 2.2 (The Future Light Cone). The upper half of the light cone comprises all points where $t > 0$ and $c^2t^2 > \vec{x}^2$. This region contains all subluminal trajectories (timelike worldlines with proper time $\tau^2 > 0$) directed forward in time. It represents the set of all future events that can be causally influenced by the event at the origin.

Definition 2.3 (The Past Light Cone). The lower half of the light cone comprises all points where $t < 0$ and $c^2t^2 > \vec{x}^2$. This region represents the history of the event at the origin, encompassing every past event that could have emitted a signal (traveling at or below the speed of light) to causally influence the present moment at O .

2.3. The 4-vector

Now that the detour is complete, we turn to the concept of a 4-vector. Recall that the most basic notion of a vector is an arrow connecting two points in space, characterised by both a magnitude and a direction:

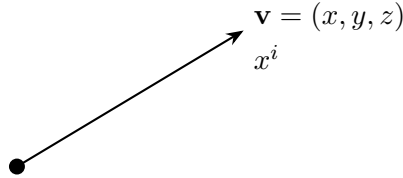


Figure 3: Representation of a spatial vector in component and index notation.

To extend the spatial vector to a 4-vector, we append a time component. This is necessary because we want to specify not only *where* an event occurs, but *when*:

$$\mathbf{v} = (t, x, y, z) = x^\mu, \quad (93)$$

where

$$x^\mu = (x^0, x^1, x^2, x^3). \quad (94)$$

Just as ordinary vectors transform under rotations, 4-vectors transform under Lorentz transformations. The index structure is simply a compact way to organise the components.

What about velocity?

In a normal vector, we would take $\frac{\Delta x}{\Delta t}$ and look at its limit. 4-dimensional velocity is a bit different.

$$U^\mu = \frac{\Delta X^\mu}{\Delta \tau} \quad (95)$$

$$(\Delta \tau)^2 = (\Delta t)^2 - (\Delta x)^2 - (\Delta y)^2 \dots \quad (96)$$

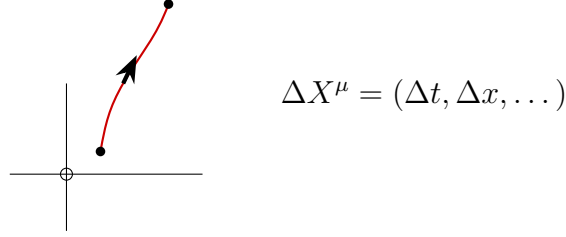


Figure 4: Spacetime path displacement and corresponding coordinate four-vector representation.

3. PARTICLE MECHANICS AND THEIR ROLE IN SPECIAL RELATIVITY

3.1. Approximations

For particles in motion, the factor $\sqrt{1-v^2} = \sqrt{1-v^2/c^2}$ appears repeatedly. When studying phenomena at very small or very large scales, it is often necessary to approximate this expression, with the quality of the approximation improving as the velocity decreases.

Recall the binomial expansion:

$$(1 + \epsilon)^p = 1 + p\epsilon + \frac{p(p-1)}{2!}\epsilon^2 + \dots \quad (97)$$

If we let $\epsilon = -v^2$ and $p = \frac{1}{2}$, then we obtain:

$$(1 - v^2)^{1/2} = 1 - \frac{1}{2}v^2 - \frac{1}{8}v^4 - \dots \quad (98)$$

This approximation becomes increasingly accurate for small, non-relativistic velocities.

The other common factor, $\frac{1}{\sqrt{1-v^2}}$, is the same expansion with $p = -\frac{1}{2}$:

$$(1 - v^2)^{-1/2} = 1 + \frac{1}{2}v^2 + \frac{3}{8}v^4 + \dots \quad (99)$$

3.2. Particles: The Big Picture

A particle can be understood as any entity that is sufficiently localised to be treated as a point — this can be essentially anything. When we refer to its position, we mean the centre of mass. Its momentum is mass times velocity, and its kinetic energy is $\frac{1}{2}mv^2$.

Several additional tools are worth keeping in mind: the principle of least action, the Lagrangian, and canonical momenta.

Definition 3.1. The **Principle of Least Action** was the notion that a physical system naturally follows the specific path, out of all conceivable paths, that minimises the action (and by *action*, we mean the total amount of energetic contributions).

The most significant concept arising from the principle of least action is the notion of a trajectory in spacetime. The trajectory of a physical particle is *timelike*.

To understand the term *timelike*, consider two arbitrary events on a spacetime diagram, described in 4-vector notation as $(\Delta t, \Delta x)$. The invariant proper time between them satisfies $(\Delta\tau)^2 = (\Delta t)^2 - (\Delta x)^2$. This is positive provided $\Delta t \gg \Delta x$, meaning there is more elapsed time than spatial separation. In this case the interval is *timelike*, and one can always find a Lorentz transformation in which the two events occur at the same spatial location (i.e. the particle has not moved).

If instead $\Delta x \gg \Delta t$, there is more spatial separation than elapsed time and the interval is *spacelike*. No Lorentz transformation can make the two events occur at the same position, though one can always find a transformation that makes them simultaneous.

3.3. Velocity

Recall that ordinary (3-dimensional) velocity is:

$$\mathbf{v}^i = \frac{d\mathbf{x}^i}{dt}.$$

The 4-dimensional velocity, however, is defined with respect to proper time:

$$U^\mu = \frac{\Delta X^\mu}{\Delta \tau}.$$

To find the relationship between the two, let

$$v^i = \frac{dx^i}{d\tau} \frac{d\tau}{dt} \tag{100}$$

$$= u^i \frac{d\tau}{dt} \tag{101}$$

We note that:

$$U^0 = \frac{dt}{d\tau} \tag{102}$$

The differential component in (101) is the inverse of (102). In differential form, we can re-write $d\tau$ as:

$$d\tau = \sqrt{dt^2 - d\vec{x}^2} \tag{103}$$

$$= dt \sqrt{1 - \frac{d\vec{x}^2}{dt^2}} \tag{104}$$

$$= dt \sqrt{1 - v^2} \tag{105}$$

This gives:

$$\frac{d\tau}{dt} = \sqrt{1 - v^2} \tag{106}$$

so that Eq. (101) becomes

$$v^i = u^i \sqrt{1 - v^2} \tag{107}$$

From Eq. (107), we can express the 4-velocity in terms of the ordinary velocity:

$$U^\mu = \frac{dx^\mu}{d\tau}, \quad U^0 = \frac{dt}{d\tau} = \frac{1}{\sqrt{1 - v^2}}, \quad U^i = \frac{v^i}{\sqrt{1 - v^2}} \tag{108}$$

3.4. The Lagrangian

Recall that in classical mechanics, the Lagrangian is:

$$\mathcal{L} = T - V \tag{109}$$

In special relativity, however, the action of a particle is purely geometric: it is proportional to the proper time elapsed along the particle's worldline in 4-dimensional spacetime. For a path between

two arbitrarily chosen events:

$$\begin{aligned}
\text{Action} &= -m \sum \Delta\tau \\
&= -m \int d\tau \\
&= -m \int \sqrt{dt^2 - \sum_i (dx^i)^2} \\
&= -m \int \sqrt{dt^2 \left(1 - \sum_i \frac{(dx^i)^2}{dt^2}\right)} \\
&= -m \int \sqrt{1 - \sum_i \left(\frac{dx^i}{dt}\right)^2} dt \\
&= -m \int \sqrt{1 - \dot{x}_i^2} dt
\end{aligned} \tag{110}$$

From Eq. (110), we can identify the Lagrangian:

$$\mathcal{L} = -m\sqrt{1 - (\dot{x}^2 + \dot{y}^2 + \dot{z}^2)}. \tag{111}$$

This Lagrangian encodes particle motion in a form valid in every reference frame.

The non-relativistic Lagrangian is:

$$\mathcal{L}_{\text{NR}} = \frac{m\dot{x}^2}{2} \tag{112}$$

The connection between Eqs. (111) and (112) is recovered by applying the binomial expansion to Eq. (111) in the limit of small velocities.

3.5. Momentum and Energy

The quantities of central interest are momentum and energy, as they are conserved and account for all particles in the universe. Recall that the momentum is given by:

$$\begin{aligned}
mu^x &= p_x = \frac{\partial \mathcal{L}}{\partial \dot{x}} \\
&= \frac{+m\dot{x}}{\sqrt{1 - v^2}} = \frac{mv_x}{\sqrt{1 - v^2}}
\end{aligned} \tag{113}$$

The same expression applies to the y and z components. Recall that momentum is conserved when the total initial momentum equals the total final momentum, or equivalently when their difference vanishes.

Note that momentum has three components. This raises a natural question: what is the fourth component $mu^0 = p_0$, and what is its physical significance? Is it energy? We can explore this via the Hamiltonian.

Recall the Hamiltonian, expressed in terms of the Lagrangian:

$$H = \sum_i \dot{x}_i p^i - \mathcal{L}, \tag{114}$$

where the momentum terms are established in Eq. (110) and the Lagrangian in Eq. (111). Substi-

tuting these into Eq. (114):

$$\begin{aligned}
 &= \sum_i \frac{m\dot{x}_i^2}{\sqrt{1-v^2}} + m\sqrt{1-v^2} \\
 &= \frac{mv^2}{\sqrt{1-v^2}} + \frac{m(1-v^2)}{\sqrt{1-v^2}} \\
 &= \frac{m}{\sqrt{1-v^2}}
 \end{aligned} \tag{115}$$

We therefore identify $mu^0 = p_0$ with the expression in Eq. (115). This is the relativistic energy, which can be re-expressed as:

$$E = mc^2 + \frac{mv^2}{2} \tag{116}$$

4. CLASSICAL FIELD THEORY

4.1. Spacetime (Review)

We recall the central objects from previous lectures. A reference frame specifies both *where* and *when* an event occurs, giving us the four-tuple

$$(x, y, z, t). \tag{117}$$

In one spatial dimension, which we freely adopt to simplify the geometry, this reduces to (x, t) . A spacetime diagram places x on the horizontal axis and t on the vertical axis. A particle moving at constant velocity v traces the worldline $x = vt$, which appears as a straight line through the origin with slope $1/v$ on this diagram.

A light ray satisfies $x = ct$, or with our conventional choice $c = 1$, it appears as a 45° line. This is important: *everything* in relativistic kinematics is anchored to the light cone.

The key tension we carried out of the last lecture is the following. If we apply the Galilean transformation,

$$x' = x - vt, \tag{118}$$

$$t' = t, \tag{119}$$

to the light ray $x = ct$, we obtain $x' = (c - v)t'$, meaning the moving observer measures a speed of light that is *less* than c . This directly contradicts Einstein's postulate. The resolution, as we saw, is that Eq. (119) is wrong: time is *not* the same in every frame. This lecture begins by asking what *replaces* the Galilean transformation.

4.2. Field

Definition 4.1. A **field** is a physical quantity that has a value at every point in spacetime. More formally, it is a function

$$\phi : \mathbb{R}^{3+1} \longrightarrow \mathbb{R} \quad (\text{for a scalar field}), \tag{120}$$

or, more generally, it may take values in some vector space or tensor space.

Fields are the central objects in classical (and quantum) field theory. Rather than tracking individual particles, we describe the state of a system by specifying $\phi(x, t)$ everywhere. The paradigmatic example from classical physics is the electric field $\mathbf{E}(\mathbf{x}, t)$: it assigns a vector to each point in space and time.

Susskind emphasises that thinking of fields as the fundamental objects is a profound shift. Whereas Newtonian mechanics begins with particles and asks how they move, field theory begins with fields distributed throughout space and asks how they evolve. The equation governing a field is typically a *partial* differential equation in x and t .

4.3. Rules

Classical field theory is built on a small number of guiding principles. Susskind lists them roughly as follows.

1. **Locality.** The equations of motion at a point (x, t) depend only on the field and its derivatives at that point. There are no instantaneous long-range influences built into the fundamental equations.
2. **Lorentz invariance.** The laws must take the same form in every inertial reference frame. This places strong constraints on which terms can appear in the action.
3. **Least action.** The physical field configuration is the one that extremises the action functional $S[\phi]$. This unifies all of the dynamics into a single object.
4. **Linearity (for free fields).** Free fields satisfy linear equations of motion, which means superposition holds: if ϕ_1 and ϕ_2 are solutions, so is $\alpha\phi_1 + \beta\phi_2$.

These principles collectively constrain the possible Lagrangian densities enormously. In practice, we write down the most general Lagrangian density consistent with the symmetries we want, and the rules above then determine the dynamics.

4.4. The Lagrangian (Field Theory)

In particle mechanics, the Lagrangian is $L = T - V$ and the action is $S = \int L dt$. For a field, we generalise this by introducing a **Lagrangian density** $\mathcal{L}$, so that

$$L = \int \mathcal{L}(\phi, \partial_\mu \phi) dx, \quad (121)$$

and the action becomes

$$S[\phi] = \int \mathcal{L}(\phi, \partial_\mu \phi) dx dt. \quad (122)$$

Here $\partial_\mu \phi$ denotes all first-order partial derivatives of ϕ with respect to the spacetime coordinates. (We will use the index notation $\partial_\mu = \partial/\partial x^\mu$ where $x^0 = t$, $x^1 = x$, etc.)

Remark 4.2. We demand $\mathcal{L}$ to be a scalar under Lorentz transformations so that S is frame-independent. This is the content of “Lorentz invariance at the level of the action.”

The simplest non-trivial scalar we can construct from a scalar field ϕ and its derivatives is

$$\mathcal{L} = \frac{1}{2}[(\partial_t \phi)^2 - (\partial_x \phi)^2] - V(\phi), \quad (123)$$

where $V(\phi)$ is a potential. The combination $(\partial_t \phi)^2 - (\partial_x \phi)^2$ is Lorentz-invariant because under a boost the t - and x -derivatives mix in exactly the right way.

4.5. Multidimensional Analogue

So far we have been working in $1 + 1$ dimensions for simplicity. In $3 + 1$ dimensions the spatial gradient $\partial_x \phi$ generalises to $\nabla \phi$, and the Lagrangian density becomes

$$\mathcal{L} = \frac{1}{2}(\partial_t \phi)^2 - \frac{1}{2}|\nabla \phi|^2 - V(\phi). \quad (124)$$

It is convenient to package the two gradient terms using the four-gradient. Define

$$\partial^\mu \phi = \eta^{\mu\nu} \partial_\nu \phi, \quad \eta^{\mu\nu} = \text{diag}(+1, -1, -1, -1), \quad (125)$$

so that

$$\partial_\mu \phi \partial^\mu \phi = (\partial_t \phi)^2 - |\nabla \phi|^2. \quad (126)$$

Then Eq. (124) reads compactly as

$$\mathcal{L} = \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - V(\phi). \quad (127)$$

The metric signature $(+, -, -, -)$ is the one Susskind uses throughout. The expression $\partial_\mu \phi \partial^\mu \phi$ is manifestly a Lorentz scalar, confirming that Eq. (127) is Lorentz invariant.

4.6. Continuum Mechanics (Motivation for Field Theory)

A helpful way to arrive at field theory is to start from the mechanics of a *discrete* chain of masses connected by springs, and then take a continuum limit.

Consider N masses, each of mass m , separated by equilibrium spacing a and connected by springs of spring constant k . Label the displacement of the n -th mass from its equilibrium position as $q_n(t)$. The Lagrangian is

$$L = \sum_{n=1}^N \left[\frac{1}{2} m \dot{q}_n^2 - \frac{1}{2} k (q_{n+1} - q_n)^2 \right]. \quad (128)$$

Now define a continuous field $\phi(x, t)$ by the identification $q_n(t) \equiv \phi(na, t)$. As $a \rightarrow 0$ with the total mass density $\mu = m/a$ and the tension $T = ka$ held fixed, the sum becomes an integral:

$$L \longrightarrow \int \left[\frac{\mu}{2} (\partial_t \phi)^2 - \frac{T}{2} (\partial_x \phi)^2 \right] dx. \quad (129)$$

We can identify the Lagrangian density as

$$\mathcal{L} = \frac{\mu}{2} (\partial_t \phi)^2 - \frac{T}{2} (\partial_x \phi)^2, \quad (130)$$

which is exactly Eq. (123) with $V = 0$ and a particular ratio of coefficients. The continuum limit is therefore a concrete *derivation* of field theory from first principles.

4.7. Generalisation

The Lagrangian density in Eq. (127) can be generalised in several ways.

1. **Multiple fields.** Replace the single scalar ϕ with a collection $\{\phi^a\}$, $a = 1, \dots, N$. Each field gets its own kinetic term, and they may be coupled through $V(\phi^1, \dots, \phi^N)$.

2. **Non-trivial potentials.** Adding a mass term $\frac{1}{2}m^2\phi^2$ gives the Klein-Gordon field (free, massive scalar). Adding $\lambda\phi^4$ gives an interacting theory.
3. **Gauge fields.** Replacing the scalar with a vector A_μ and demanding local symmetry yields electromagnetism. The Lagrangian density is

$$\mathcal{L}_{\text{EM}} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu}, \quad F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu. \quad (131)$$

4. **Coupling to matter.** One can add terms coupling the field to a current J^μ , for instance $\mathcal{L}_{\text{int}} = J^\mu A_\mu$ for electromagnetism. This is how Maxwell's equations with sources arise from the action.

The remarkable power of the Lagrangian formalism is that all of these distinct-looking theories are unified by the *same* variational principle applied to different $\mathcal{L}$.

4.8. Equations of Motion

We derive the equations of motion by extremising the action $S[\phi]$ in Eq. (122). Consider a small variation $\phi \rightarrow \phi + \delta\phi$ with $\delta\phi = 0$ on the boundary. Expanding to first order,

$$\delta S = \int \left[\frac{\partial \mathcal{L}}{\partial \phi} \delta\phi + \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \delta(\partial_\mu \phi) \right] dx dt. \quad (132)$$

Integrating the second term by parts and using $\delta\phi = 0$ at the boundary,

$$\delta S = \int \left[\frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \right) \right] \delta\phi dx dt. \quad (133)$$

For $\delta S = 0$ for all $\delta\phi$, we require the integrand to vanish pointwise. This gives the **Euler–Lagrange equation for a field**:

$$\boxed{\partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \right) - \frac{\partial \mathcal{L}}{\partial \phi} = 0.} \quad (134)$$

Applying Eq. (134) to the free Lagrangian density in Eq. (127) with $V = \frac{1}{2}m^2\phi^2$:

$$\frac{\partial \mathcal{L}}{\partial \phi} = -m^2\phi, \quad (135)$$

$$\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} = \partial^\mu \phi, \quad (136)$$

so Eq. (134) becomes

$$\partial_\mu \partial^\mu \phi + m^2\phi = 0 \quad \implies \quad \square\phi + m^2\phi = 0, \quad (137)$$

where $\square = \partial_t^2 - \nabla^2$ is the d'Alembertian. Equation (137) is the **Klein–Gordon equation**, the fundamental wave equation for a massive scalar field.

4.9. Wavelike Motion

Setting $m = 0$ in Eq. (137) gives the **wave equation**:

$$\square\phi = \partial_t^2\phi - \nabla^2\phi = 0. \quad (138)$$

In one spatial dimension, Eq. (138) becomes

$$\frac{\partial^2 \phi}{\partial t^2} = \frac{\partial^2 \phi}{\partial x^2}, \quad (139)$$

(where we set $c = 1$). The general solution is

$$\phi(x, t) = f(x - t) + g(x + t), \quad (140)$$

for arbitrary functions f and g . These represent a right-moving wave and a left-moving wave respectively, both travelling at speed $c = 1$. Equation (140) is d'Alembert's solution.

For a plane wave ansatz $\phi = e^{i(kx - \omega t)}$, substitution into Eq. (138) gives the **dispersion relation**:

$$\omega^2 = k^2 \implies \omega = \pm k. \quad (141)$$

For the massive Klein–Gordon equation the dispersion relation becomes

$$\omega^2 = k^2 + m^2, \quad (142)$$

which is recognisable as the relativistic energy-momentum relation $E^2 = p^2 c^2 + m^2 c^4$ in natural units. This is not a coincidence: it reflects the deep connection between quantum field theory and special relativity.

4.10. Special Relativity and Lorentz Invariance of the Action

We now close the loop between classical field theory and special relativity. Recall the Lorentz transformation:

$$x' = \gamma(x - vt), \quad (143)$$

$$t' = \gamma\left(t - \frac{v}{c^2}x\right), \quad (144)$$

where $\gamma = (1 - v^2/c^2)^{-1/2}$ is the Lorentz factor. Setting $c = 1$,

$$\gamma = \frac{1}{\sqrt{1 - v^2}}. \quad (145)$$

The key object invariant under Eq. (143)–(144) is the spacetime interval

$$ds^2 = dt^2 - dx^2, \quad (146)$$

which is unchanged by a Lorentz boost. The action in Eq. (122) is a spacetime integral of a Lorentz scalar, so it is manifestly invariant. This means the equations of motion (Eq. (134)) take the same form in every inertial frame, confirming that classical field theory is fully consistent with special relativity.

A very direct way to see this is to note that the d'Alembertian $\square = \partial_t^2 - \nabla^2$ can be written as

$$\square = \partial_\mu \partial^\mu = \eta^{\mu\nu} \partial_\mu \partial_\nu, \quad (147)$$

which is a Lorentz scalar operator. Any equation built from $\square$ acting on a scalar field is therefore Lorentz covariant. The Klein–Gordon equation $\square\phi + m^2\phi = 0$ is the simplest example of a Lorentz-invariant equation of motion for a field.

Summary. This lecture established the machinery of classical field theory from the ground up. We introduced the concept of a field, stated the governing principles (locality, Lorentz invariance, least action), motivated the Lagrangian density through the continuum limit of a discrete chain, derived the Euler–Lagrange equations for fields, identified the Klein–Gordon and wave equations as the simplest equations of motion, and finally verified that the whole framework is Lorentz-invariant. The natural next step is to apply this machinery to specific physical fields — electromagnetism being the most important classical example.

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